Lab-2 Ex2 - Quadratic Equations (PreLab*)

> This is a PreLab exercise. Please do the PreLabs after lecture and before you come to lab!

It is customary to have quadratic equation solvers as examples when you first learn any programming languages. A ***discriminant*** of a quadratic equation with ***coefficients*** a, b and c is defined as:

$$\Delta = b^2 - 4ac$$

Below you will find a working C program that calculates the ***discriminant*** of a quadratic equation with ***coefficients*** a, b and c:

```c
#include <stdio.h>

int main(void) {
  double a, b, c, discriminant;

  printf("a = ");
  scanf("%lf", &a);
  printf("b = ");
  scanf("%lf", &b);
  printf("c = ");
  scanf("%lf", &c);

  discriminant = b * b - (4 * a * c);
  printf("discriminant is %.3f\n", discriminant);

  return 0;
}
```

As a warm-up, please type in the program above accurately so that it will correctly calculate the ***discriminant***. Note the structure of the C program and the spacing, called **indentation**, in the example. Try to follow the style in all lab exercises!

If your program is correct, it will match with the following sample behaviour:

Sample Run #1 (user input is bolded and in italic)

a = ***1***

b = ***2***

c = ***4***

discriminant is -12.000

Sample Run #2 (user input is bolded and in italic)

a = ***0.5***

b = ***3***

c = ***0.33***

discriminant is 8.340